

The Future of Embodied Interaction in the Age of Disembodied Intelligence

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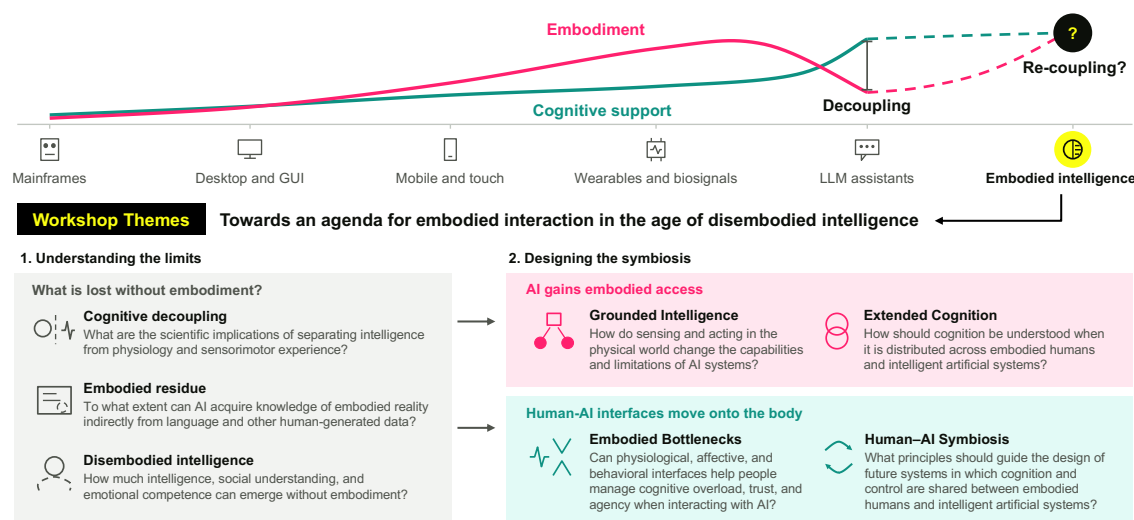


Fig. 1. Core Concepts and Open Questions for Embodied and Disembodied AI Research.

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Human-computer interaction has advanced through three complementary forces: (1) increasing cognitive support, (2) increasing embodiment, and (3) increasing integration, transforming computers into continuous partners. Recent advances in Large Language Models (LLMs) accelerate the first and third by providing unprecedented cognitive support and becoming active participants in human thought, yet largely neglect the second. While wearables and physiological interfaces have brought computation closer to human perception, action, physiology, and emotion, today's most capable AI systems remain disembodied. They increasingly participate in communication, learning, emotional support, and decision-making without direct access to the Umwelt that grounds human experience. This decoupling of intelligence from embodiment raises fundamental questions about cognition, empathy, trust, agency, and human-AI interaction. What does this mean for future interfaces? Our workshop brings together researchers from multiple HCI disciplines to examine what is lost without embodiment and how intelligence and embodiment might be redesigned and re-coupled toward human-AI symbiosis.

CCS Concepts: • **Human-centered computing** → **Interaction design theory, concepts and paradigms**; • **Computing methodologies**; • **Security and privacy**;

Additional Key Words and Phrases: Human-AI interaction, embodiment, physiological computing, affective computing

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1 Motivation

1.1 Computing's Movement Toward the Body

The history of computing shows a progressive integration of technology into human cognitive and bodily processes. The closer an interface sits to these processes, the more continuously it can adapt to users and their context, so that support no longer depends on explicit, on-demand interaction [21, 29]. Decades of work in human-computer interaction, affective computing, and physiological computing have pursued this path by incorporating perception, action, emotion, and physiology into interactive systems [17, 26], ranging from long-standing efforts to read bodily proxies of emotion [6, 24, 35, 36] to wearable systems that turn behavior and physiological signals into everyday interfaces [11, 13, 16]. Conceptually, this repositions the body, which is no longer only a channel for input but a continuous source of context through which systems interpret and support human activity, an interpretation that aligns interface design with accounts of embodied and extended cognition [8]. These developments have produced increasingly tight forms of human augmentation [34], especially in human-computer integration [12, 20, 22]. Frameworks like the JIZAI Body reimagine the simultaneous control of natural and artificial body parts [15], and trends to implement interfaces beyond realism have also highlighted the need to consider sensorimotor integration and users' internal models to achieve body transformations in Virtual Reality [1, 37]. Taken together, multiple working definitions of embodiment have emerged. In robotics, embodiment means that body shape, materials, sensors, actuators, and their dynamics perform part of the information processing required for intelligent behavior, and physical instantiation is a requirement for intelligence because bodily morphology and properties constrain and enable cognition rather than merely carrying out commands from a controller [25]. From a phenomenological point of view, embodiment is the lived, perceptual engagement of the body with the world. The body is not merely an object we possess; it is the standpoint through which the world is experienced [27]. Cognition depends on experiences arising from a body with particular

sensorimotor capacities, which exist within biological, psychological, and cultural contexts. In this sense, cognition emerges through recurrent sensorimotor patterns and bodily action [31].

1.2 Cognitive Support Enabled by LLMs

In contrast to this previous trajectory, recent advances in Large Language Models (LLMs) represent a major shift. Unlike previous technologies that primarily stored or transmitted information, LLMs actively participate in cognitive processes by generating explanations, summarizing information, fostering creativity, supporting decision-making, and providing emotional support [5, 18, 19, 23, 28, 39]. As a result, humans increasingly rely on AI systems as cognitive interaction partners rather than passive tools [9, 33]. LLMs can therefore be viewed as a new form of cognitive technology that operates directly on language, the medium through which humans organize and extend thought. But human thinking is not carried by language alone. Work in psychology and cognitive neuroscience shows that cognition, emotion, and motivation are grounded in sensorimotor and bodily experience, not in symbol manipulation alone [4, 38]. Expertise develops in the same manner: Dreyfus and Dreyfus [10] describes how skilled performance moves from following explicit rules to acting on contextual, embodied intuition. Therefore, support given in language alone reaches only part of what it addresses, and the most capable cognitive technology to date has advanced along a different path from interaction with the body.

1.3 The Tension Between Embodied Interaction and Disembodied Intelligence

Two paths have thus developed side by side. Interaction research has moved technology ever closer to the body, while AI has advanced by remaining linguistic and screen-based, without a lived sensorimotor or interoceptive loop. This dominant interaction mode of current foundation models creates a fundamental tension across multiple scientific disciplines [2]. Modern AI systems have achieved unprecedented capabilities while remaining fundamentally disembodied. They possess no physiology, sensorimotor experience, or direct bodily grounding, yet participate in activities traditionally associated with embodied processes such as intelligence, empathy, communication, and social interaction. This stands in contrast to earlier work on embodied conversational agents, which explicitly incorporated bodily and nonverbal behavior into human-agent communication [3]. Likewise, research on virtual co-embodiment shows that bodily interaction can shape experiences of agency, body ownership, and co-presence [32]. Recent work has begun to bridge these trajectories by embedding generative AI within embodied interaction, for example: enabling the typically disembodied LLM assistance to instead become physical assistance (i.e., act in the body rather than audiovisually) [14]; or even integrating LLM-mediated speech, voice cloning, and avatar embodiment to explore how AI-generated expression can become experienced as an extension of the self [40]. Yet work spanning AI and cognitive neuroscience makes this gap explicit, arguing that even multimodal models lack the interoceptive monitoring that humans use to anchor abstract concepts in physical experience [7]. This decoupling of intelligence from embodiment challenges long-standing assumptions about embodiment, despite the need to integrate technological interfaces within the human body [21]. Therefore, AI augmentations should consider the embodiment of the AI itself, the embodiment of the human-facing interface, and the allocation of authority between person and system when humans and AI interact within the same or different bodies. Investigating what happens when increasingly disembodied forms of intelligence are re-coupled with human embodiment through interfaces is thus a pressing topic for the CHI community.

2 Core Concepts and Open Questions

We will tackle two questions: (1) *what is lost when intelligence is decoupled from embodiment*, and (2) *how might embodiment be ethically reintroduced as we design future human–AI symbiosis?* (Fig. 1). As a solution, AI systems could be given greater embodied access to the world, and human-facing interfaces could move closer to the body. These strands are not interchangeable: giving a system its own senses and putting the interface on the body raise different questions about grounding, privacy, and control.

3 Organizers

Monica Perusquia-Hernández (<https://www.monicaperusquia.com/>) is an assistant professor at the Nara Institute of Science and Technology, Japan, working on affective and physiological computing, embodiment, and human–AI interaction, with a focus on wearable sensing and AI-mediated communication. She has held leadership roles at IEEE FG and ACII, and has co-organized workshops at CHI, MobileHCI, ACII, and other local venues in Japan.

Michael T. Knierim (<https://michaelknierim.info/>) is a research group head at the Karlsruhe Institute of Technology, Germany. He works on wearable neurotechnology and physiological computing, investigating both the demands of digitally mediated knowledge work, such as mental workload and fatigue, and its positive experiences, such as flow and engagement. He has served on the program committees of ISWC and the Workshop on Open Wearables.

Elisabeth André (<https://elisabethandre.de/>) is a Professor of Human-AI Interaction at the University of Augsburg, Germany. She works on human-centered AI, affective computing, and intelligent user interfaces, with a focus on embodied conversational agents and social signal processing. She is a member of the ACM SIGCHI Academy, served as General Chair of ICMI 2023 and ICMI 2027, and received the 2021 Gottfried Wilhelm Leibniz Prize, Germany’s most prestigious research award.

Ekaterina R. Stepanova is a Banting Postdoctoral Fellow in the Interaction Design Group at the KTH Royal Institute of Technology, Sweden. Drawing on cognitive science and soma design, she develops Intercorporeal Design, a program for eliciting embodied social interactions through technology and for examining how embodied relations emerge between people and artificial agents. Her work spans XR, biosignals, soft robotics, and embodied AI.

Steeven Villa (<https://posthci.com/>) is a researcher at LMU Munich, Germany, and Senior Neuroscience CX Researcher at T-Systems Iberia. He works on human augmentation, developing technologies that extend cognitive, motor, and sensory capabilities and studying their effects on cognition, behavior, and experience. His work spans virtual and augmented reality, wearables, electrical muscle stimulation, and electroencephalography.

Pedro Lopes (<https://plopes.org/>) is an Associate Professor of Computer Science at the University of Chicago, where he directs the Human-Computer Integration Lab and is affiliated with the Neuroscience Institute. He develops interfaces that operate through the body, combining haptics, wearables, and immersive systems to study and augment movement, perception, and agency [20, 22], most recently in embodied-AI systems that give context-sensitive physical assistance while preserving human control [14]. He served as Technical Program Chair for CHI 2026 and organizes research communities around NeuroHCI [30].

Mar Gonzalez-Franco (<https://research.google/people/108218/>) is a computer scientist and neuroscientist at Google, United States, where she leads the BIRD lab on Blended Intelligence Research and Devices. She works on immersive technologies and avatar embodiment, studying how virtual bodies shape perception, behavior, and the sense of self. She received the 2022 IEEE VGTC VR New Researcher Award and was named an NAE Frontiers Engineer.

Albrecht Schmidt (<https://uni.ubicomp.net/as/>) holds the chair in Human-Centered Ubiquitous Media at LMU Munich, Germany, where he is Dean of the Faculty of Mathematics, Computer Science and Statistics. He works on how digital technologies amplify human cognition and perception, spanning human-centered AI, ubiquitous computing, AR/VR, and physiological sensing. He introduced the notion of implicit human-computer interaction and leads the ERC Advanced Grant AI-Twin.

Abdallah El Ali (<https://abdoelali.com>) is a research scientist at Centrum Wiskunde & Informatica (CWI) in Amsterdam, and Assistant Professor at Utrecht University. He leads research in Affective Interactive Systems and Trustworthy Human-AI Interaction, with a focus on human augmentation, AI self-extension across physical and digital environments, and AI transparency. He serves as Chair of CHI Nederland, the ACM SIGCHI Netherlands Chapter.

4 Activities

Plan To Publish Proceedings. We will publish the workshop proceedings on CEUR-WS.org. Moreover, the workshop website will host papers accepted two weeks before the conference upon the authors' consent.

Website. The website will contain the call for papers, links to materials and activities for asynchronous engagement, and links to the accepted extended abstracts.

Accessibility. We are committed to fostering an inclusive workshop environment that welcomes diverse perspectives and enables equitable participation. We would like to keep CHI's standard on the use of accessibility tools. To facilitate organization, we will inquire about accessibility needs once the attendee list is available.

Post-Workshop Plans. We will create a summary of guidelines and future directions for embodied human-AI interaction and publish it on the workshop website as living documentation of the workshop's outcomes. The collection of outcomes would also include a list of resources provided by the participants to their own tools, such as hardware, software, and interesting papers to share. These will serve as a basis for a joint publication by organizers and participants, including a taxonomy of embodied-disembodied interaction levels, and a joint position paper (e.g., in *Communications of the ACM*). Finally, we will prepare a proposal for a special issue at a journal (e.g., TOCHI) on new (embodied) interaction paradigms with AI, potentially including a Shonan or Dagstuhl proposal and further workshops at SIGCHI conferences.

Workshop Structure. In two 90-minute sessions, we aim to inspire and encourage discussion (Table 1). We will collect insights from the discussions and the materials generated during the workshop for a follow-up paper to which all participants will be invited to co-author. Each step is designed to function as inspiration and input to the next step. Through live documentation of each step's outputs, we will ensure that information flows to the next step, even if the attendees of each session differ. First, the submitted abstracts will serve as inspiration to start the discussion on the biggest challenges to tackle and to envision how they could be solved in 10 years. These visions will serve as the starting point for more detailed proposals, and the closing fishbowl will provide a discussion and critique of the proposed ideas. The workshop is planned for approximately 20 people. For larger numbers of attendees, we will escalate by using participation and voting digital tools to gather and summarize information in real time. Before and during the workshop, live documents posted on the website will be made available to facilitate preparation, discussion, and documentation. Additionally, we will create a Slack or Discord Server to serve as a platform for engaging participants, continuing the evolving discussion of the challenges and opportunities presented by designing for embodiment with AI, posting questions, sharing posts about their work, recruiting participants, and forming collaborations.

Table 1. Schedule of the two-session workshop

Session	Duration	Activity
1 (90')	5'	<i>Introduction.</i> Workshop organizers, goals, methods, and post-workshop events will be explained.
	5'	<i>Embodied Warm-Up.</i> As an ice-breaker and to root participants in their embodied experiences, we will facilitate a contact improvisation-inspired movement exercise where participants will experience how a sense of agency can be transferred between collaborating agents within a symbiotic coupling
	50'	<i>Mapping the Landscape.</i> The state of the art will be explored through lightning talks (3' each + 2' questions x 10) based on selected position papers submitted to the workshop.
	5'	<i>Landscape theme aggregation.</i> Workshop organizers will present a thematic analysis of the lighting talks, prepared in advance from the workshop submissions.
	25'	<i>Grand Challenge Wall.</i> Attendees will propose grand challenges; then, these will be grouped into "Future Headlines of Embodied Human-AI Interaction (2037)". Post-its or a digital board tool will be used.
2 (90')	30'	<i>Designing the Future Agenda.</i> Attendees will group around the five popular headlines elaborated in session 1. Then they will create moonshot proposals for hypothetical projects to achieve the headline.
	25'	<i>Presentation of moonshot proposals.</i> Attendees will share their proposed idea.
	5'	Break
	30'	<i>Closing fishbowl.</i> A structured open discussion to answer "What should this field prioritize in the next decade to optimize embodied throughput/communication between people and their AIs?" This is intentionally broad; disagreement is welcome. The discussion happens as a dynamic panel in which a small <i>inner circle</i> of speakers discusses a shared question while the audience observes. Any attendee may tap a participant on the shoulder to swap in. It will be opened by the organizers and then evolve through free participation.

5 Call for Participation

HCI has strived to (1) increase cognitive support; (2) increase embodiment; and (3) increase integration. Large Language Models (LLMs) accelerate the first and third trends by providing unprecedented cognitive support and becoming active participants in human thought processes. Yet humans remain embodied while AI becomes increasingly disembodied. What does this mean for future interfaces? We seek to understand the future role of embodiment in human-AI symbiosis and define the theoretical foundations, design principles, and technological opportunities for embodied interaction in the age of disembodied intelligence. We invite two-page extended abstracts that present a project or articulate a challenge related to this call, explicitly mentioning how it relates to the core concepts and open questions of this workshop: Disembodied Intelligence, Embodied Residue, Cognitive Decoupling, Grounded Intelligence, Extended Cognition, Embodied Bottlenecks, and Human-AI Symbiosis. Previously published papers summarized as extended abstracts are also welcome. Submissions should be sent to EasyChair ¹ and must adhere to accessibility and AI usage guidelines outlined by ACM and use the CEUR Workshop Proceedings template. Submissions will be selected based on the quality and contribution of the work relating to the workshop themes, with a special focus on novel interaction paradigms with AI. Abstracts with complementary relevance will be invited for lightning talks. All abstracts will be summarized by the workshop organizers, made available on the workshop's website, and published as CEUR proceedings. At least one author of each accepted submission must attend in person and register for the workshop and the CHI'27 conference. For more information, please visit <https://disembodied-intelligence-chi.github.io/>.

¹<https://easychair.org/conferences/?conf=eiadi2027>

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